



Correction: Agile meets quantum: a novel genetic algorithm model for predicting the success of quantum software development project

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Correction to: Automated Software Engineering (2024) 31:34
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In this article, the Appendix section was missed during production processing from this article and should have read as follows:

The original article can be found online at <https://doi.org/10.1007/s10515-024-00434-z>.

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Appendix A: Categories, challenges, causes and their respective best practices.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
Knowledge and awarness	Ch1 (Knowledge gap)	C1 (Lack of domain specific knowledge) [BP1], [BP2], [BP3], [BP4], [BP8], [BP10]	BP1-Foster collabora-tion and camaraderie through structured activities, aiming to enhance interpersonal relationships and team synergy.
	Ch2 (Team dynamics)	C2 (Lack of market intrest) [BP2], [BP5], [BP6], [BP7], [BP9]	BP2-Simplify complex quantum development concepts into digestible content to improve team comprehension and application skills.
	Ch3 (Quantum software development educa-tion)	C3 (Limited acadamic reserch) [BP2], [BP5], [BP8], [BP10]	BP3-Organize regular workshops and training sessions to raise the team's awareness and proficiency in quantum software development.
		C4 (Funidng sources) [BP6], [BP7], [BP8], [BP9]	BP4-Implement strategies and exercises designed to solidify trust and unity among team members, crucial for tackling sophisticated projects. BP5-Educate teams on the extended timelines often associated with quantum develop-ment and its potential long-term technological impact.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
Sustainable scaling	Ch4 (Ethically aligned quantum software development)	C5 (Complex technical requirements) [BP13], [BP14], [BP15], [BP16]	BP6-Upgrade and refine the technological infrastructure to support the unique demands of quantum software development effectively.
			BP7-Increase awareness within the industry about the benefits and applications of quantum software, fostering broader acceptance and investment.
			BP8-Provide support and resources for executing quantum software engineering research in academic settings to push the boundaries of the field.
			BP9-Develop a deep understanding of how quantum software systems will interact with and affect business operations and strategies.
Sustainable scaling	Ch5 (Agile-quantum ecosystem)	C6 (Cross-disciplinary integration difficulties) [BP12], [BP15], [BP16], [BP17]	BP10-Stay informed about and integrate domain-specific best practices to ensure excellence and relevance in quantum software development.
			BP11-Develop agile frameworks that incorporate ethical guidelines specifically designed for quantum software development, ensuring that products are aligned with societal values and norms.
Sustainable scaling	Ch5 (Agile-quantum ecosystem)	C6 (Cross-disciplinary integration difficulties) [BP12], [BP15], [BP16], [BP17]	BP12-Form agile teams comprising experts from various disciplines to facilitate the integration of diverse knowledge, which is crucial for the complex nature of quantum software development.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
		C7 (Rapid pace of innovation) [BP11], [BP12], [BP13], [BP14], [BP15]	BP13-Establish innovation labs that operate within the agile methodology, dedicated to exploring and rapidly prototyping with emerging quantum technologies. BP14-Implement continuous learning and development programs within agile teams to keep pace with the rapid innovation in quantum computing. BP15-Create protocols that guide the effective integration of agile practices within the quantum software development process, ensuring that agility and quantum innovation complement each other. BP16-Align quantum software development projects with Sustainable Development Goals to ensure that scaling up quantum computing capabilities does not come at the expense of environmental or social sustainability. BP17-Introduce governance structures that apply agile principles to manage the complexity and ethical considerations of quantum software projects, allowing for flexibility and adaptability in decision-making processes.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
Quantum-aware tools and technologies	Ch6 (Classic-quantum tailoring)	C8 (Right tool for right job) [BP18], [BP20], [BP24]	BP18-Develop and integrate agile toolkits that are tailored for quantum software development. These should facilitate the classic-quantum tailoring process and ensure that developers have the right tools for quantum-specific tasks.
	Ch7 (Continuous SE infrastructure)	C9 (Lack of industrial interest) [BP21], [BP23], [BP25]	BP19-Implement agile practices to regularly evaluate and select quantum development tools. This iterative tool assessment should be part of the sprint reviews, ensuring continuous alignment with project needs.
		C10 (Limited resources) [BP19], [BP21], [BP22], [BP24]	BP20-Within the agile framework, allocate resources specifically for exploring and integrating quantum technologies. This addresses the limited resources challenge by earmarking investments for quantum development. BP21-Form interest groups or committees within agile teams that focus on increasing industrial interest in quantum software development through demonstrations, publications, and industry partnerships. BP22-Designate specific agile sprints to focus on the transformation process from legacy software to quantum software. These sprints would deal with refactoring, integration, and the adoption of new quantum technologies.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
Standards & specifications	Ch8 (Processes standardization)	C11 (Varying Interpretations of Agile Methodologies) [BP29], [BP32]	BP23-Conduct workshops to train agile teams on how to effectively manage the technological paradigm shift, incorporating quantum computing concepts into their existing agile practices.
			BP24-Develop CI/CD pipelines that are capable of supporting both classical and quantum software development, ensuring that agile practices can sustain a continuous SE infrastructure.
			BP25-Encourage agile teams to engage in technological foresight to anticipate and prepare for future shifts in quantum technology, integrating this into their sprint planning and backlog refinement.
	Ch9 (Optimum documentation)	C12 (Rapid Evolution of Quantum Technologies) [BP27], [BP28], [BP31], [BP32], [BP34]	BP26-Develop maturity models specific to agile-quantum development to assess and guide teams on their journey towards optimized agile practices in quantum software creation. BP27-Institute standard assessment protocols that evaluate the effectiveness of agile practices and documentation in quantum development, ensuring they meet the high-quality standards required in this field.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
		C13 (Complexity of Quantum Computing Concepts)[BP29], [BP31],[BP33]	BP28-Create a comprehensive integration framework that lays out best practices for merging agile methodologies with quantum software development, including checklists and templates for standardized processes.
		C14 (Necessity of Integration with Existing Standards)[BP26], [BP27],[BP28], [BP30],[BP31]	BP29-Implement quality standards for documentation that are specifically designed for the quantum software domain, guiding teams on creating clear, concise, and useful documentation artifacts. BP30-Integrate sprints that are dedicated to ensuring the project's compliance with established frameworks and standards, allowing teams to correct course and align with best practices throughout the development cycle. BP31-Encourage the use of SDKs that come with built-in standards and documentation guidelines, making it easier for agile teams to adhere to best practices without extensive overhead. BP32-Compile playbooks that detail the agile-quantum processes, offering standardized procedures and documentation templates that can be customized for various quantum development scenarios.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
Common causes accross all the categorieis		C15 (Technological Paradigm Shift) [BP35], [BP38], [BP40], [BP41]	BP33-Conduct focused workshops to train agile teams on creating tailored documentation that meets the specific needs of quantum software development while adhering to agile principles. These workshops would emphasize the creation of just-enough and just-in-time documentation, ensuring relevance and agility.
			BP34-Dedicate specific sprints to the development and refinement of documentation, ensuring that it keeps pace with the rapid development cycles typical in quantum software projects. These sprints would focus on updating documentation to reflect new learnings, architectural decisions, and code changes, maintaining an optimum level of documentation throughout the project lifecycle.
Common causes accross all the categorieis		C15 (Technological Paradigm Shift) [BP35], [BP38], [BP40], [BP41]	BP35-As quantum computing is a technological paradigm shift, it is essential to invest in continuous learning opportunities for team members to keep up with the latest quantum computing developments and practices. Regular training sessions, workshops, and seminars can help bridge the gap between research and practice.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
		C16 (Transforming process from legacy to quantum software) [BP35], [BP36], [BP38]	BP36-When moving from legacy to quantum software, employ an incremental approach that allows for iterative testing, validation, and integration of quantum components.
		C17 (Gap between research and practice) [BP35], [BP37], [BP40]	BP37-Form alliances with academic and research institutions to stay abreast of the latest quantum computing research and to help translate those findings into practical applications.
		C18 (Emerging working culture) [BP41], [BP42], [BP43]	BP38-Implement an agile methodology that is flexible enough to accommodate the rapid changes inherent in quantum technology and allows for adapting workflows to integrate new findings and techniques.
		C19 (Interdisciplinary Collaboration Barriers) [BP39], [BP40], [BP42]	BP39-Create cross-functional teams with diverse skill sets, including quantum physicists, software developers, and domain experts, to foster interdisciplinary collaboration and innovation. BP40-Develop internal knowledge-sharing platforms to disseminate quantum computing knowledge and practices within the organization, bridging the gap between different disciplines.

Categories (Khan et al. 2023)	Concepts (challenges(Ch)) (Khan et al. 2023)	Causes	Best practices (BP)
			BP41-Encourage a culture that embraces experimentation and tolerates failure, recognizing that the quantum field is exploratory and that failure can provide valuable insights.
			BP42-Utilize modern communication and collaboration tools to support the emerging working culture, which may be more distributed and remote, particularly in the interdisciplinary field of quantum computing.
			BP43-Employ agile project management techniques specifically tailored for quantum software development projects, accounting for their high uncertainty, rapid evolution, and the need for close collaboration across different fields.

The original article has been corrected.

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